# ASSIGNMENT 2, FINANCIAL ECONOMETRICS

**This assignment covers the materials discussed in class. Submit it on Canvas. Follow the instructions, including the notes at the end of this document. Some of you may want to work individually. However, you can work in groups of up to three students.**

## 1. Data

The analysis uses the time series listed below to perform ARIMA and Seasonal ARIMA models. These series relate to bank financial performance, specifically credit card portfolio balances and delinquencies. The R code included with the assignment connects to the data source and downloads the following time series.

*a. Delinquency Rate on Credit Card Loans for Commercial Banks*

This series is seasonally adjusted, so it does not require seasonal modeling.

*b. Credit Card Balance for Commercial Banks*

This series is not seasonally adjusted. It will require seasonal modeling.

## 2. Code

The assignment will require using R to generate ARIMA and SARIMA forecasts. The R program "CCDQSeries", attached to the assignment, downloads the data and fits an ARIMA model. Similarly, the R program "CCBALSeries", attached to the assignment, downloads data and fits a SARIMA model.

## 3. Analysis

As noted above, you will run the R code to produce non-seasonal and seasonal models, including various outputs, and the resulting forecasts. The assignment requires you to run the code and answer related questions. The analysis will involve the following:

*a. Description of time series*

- You will use the provided report template to paste charts and outputs obtained from the CCDQSeries and CCBALSeries programs. You will also discuss the ADF, ACF, and PACF outputs as required.

*b. ARIMA outputs and forecasts on the non-seasonal series*

- You will produce various outputs and the forecast using the ARIMA technique. Generate the forecasts and offer observations based on the R outputs.

c. *SARIMA outputs forecasts on the seasonal series*

- You will produce various outputs and the forecast using the Seasonal ARIMA technique. Generate the forecasts and offer observations based on the R outputs.

*d. Discussion Questions*

- You will answer discussion questions as presented on the report template.

## 4. Report Template

Summarize the outputs and analysis in the attached report template. Text and output boxes are provided. Include commentary and results as instructed on the report template. Assume you are preparing the report in a professional setting, for colleagues or supervisors. When you complete the analysis, save it as a PDF document.

**Other Notes**

- **Assume you are preparing this report for presentation to a group of colleagues or supervisors. Use the provided template and maintain a professional report style. Boxes are provided for convenience and to keep materials organized. You can delete the boxes and paste the text and outputs into the Word document. Either way, provide all the information requested on the report template.**

- **Be clear and organized when discussing the results or answering the questions. Clarity and organization will factor into the evaluation.**

- **As noted earlier, when answering the discussion questions, you may cite papers, books, or other documented sources to support your explanations or arguments (if you consider it necessary). Copying and pasting materials directly from a search engine or using AI tools to prepare answers is prohibited and violates the JHU student conduct code. The instructor may use tools to identify content obtained through browser or artificial intelligence sources.**

- **Submit the report in PDF format. Canvas works better with PDF files. You only need to submit your PDF file. Do not submit CSV or R files.**

- **You can work in groups of up to three students. Submit only one report per group. If you work in groups, ensure all student names are on the report cover.**

- **Whether you work individually or as a team, do not share outputs or answers to the discussion questions with other students. If you work individually, keep the solutions in your individual report. If you work as a team, prepare the team's solutions and record them in your team report. Again, submit only one report for your team, with all team members' names on the PDF report cover.**

- **Start early to have sufficient time to assemble the materials, including the answers to the discussion questions. Putting the report together will take time.**

- **The assignment follows similar steps as for Assignment 1. Run the code, paste the required outputs, comment on them, and address the discussion questions.**